NABU COMMAND CENTER

VERSION 1.0

USER MANUAL & OPERATOR GUIDE

Atomic Clock • Local Weather • Earthquakes • Space Weather • ISS • Airspace • Task Manager • Music

Owner-captured physical NABU running the final accepted Airspace / ADS-B view. Visible build: NCC-SPLASH-260826-LV2.

Created by Derek Leger aka (Super_Derek)

Production User Manual • 26 August 2026

Document Control

Field	Current manual basis
Product	NABU Command Center v1.0
Manual purpose	End-user and operator instructions for the accepted v1.0 feature set
Software baseline	NCC-SPLASH-260826-LV2
Artifact size	000001.bin and 000001.NABU: 59,501 bytes each
SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
MAME status	OWNER VERIFIED
Physical NABU status	OWNER VERIFIED
Repository status	Owner-accepted hardware-verified release-candidate working tree; final LV2 commit/tag freeze pending
Creator credit	Derek Leger aka (Super_Derek)
Primary engineering reference	NABU Command Center Master Production Engineering White Paper v1.1

	RELEASE IDENTITY Always check the BUILD_ID shown by the running program before troubleshooting. This manual documents NCC-SPLASH-260826-LV2. Runtime acceptance and a Git release freeze are separate evidence gates.

	PHYSICAL DEPLOYMENT BOUNDARY The exact LV2 binary has run successfully on physical NABU hardware. The complete third-party physical interconnect manifest (specific interface model, COM port, cable/pinout, and serial settings) is not yet part of the frozen documentation. Do not infer those details from the MAME configuration.

How to Use This Manual

This is the operator manual. It explains what the Command Center displays, what every normal key does, how to start the Gateway/Adapter/NABU chain, how to use every v1.0 module, and how to diagnose common failures without turning a user problem into a source-code experiment.

For source code, byte-level record layouts, linker/memory accounting, build provenance, and ground-up engineering reconstruction, use the Master Production Engineering White Paper v1.1.

Contents

PART I — Quick Start and Dashboard Tour

1. What NABU Command Center Is

NABU Command Center is a native Z80 situational-information console for the NABU Personal Computer. A modern Gateway gathers current Internet data, validates and reduces it, then publishes small bounded records through the NABU Internet Adapter. The NABU remains the interactive console: it validates records, preserves last-valid state, draws the interface, reads the keyboard, advances the clock, and produces AY sound.

Figure 1. User-level architecture. The Gateway absorbs modern web/API complexity; the NABU receives bounded prepared state.

2. Version 1.0 Feature Set

Area	What v1.0 provides
Atomic Clock	Network-maintained time with locally advancing display between refreshes.
ZIP / Location	Five-digit ZIP entry, dynamic city/state label, and ZIP-centered local context.
Local Weather	Current temperature, condition, wind, pressure, trends/history, and NWS alert summary.
Earthquake	Local and Global modes, recent event selection, magnitude/depth/age/region, graphical terrain/vector presentation.
Space Weather	NOAA/SWPC current values plus Kp and solar-wind trend visualization.
Satellite / ISS	Live ISS identity and position state with native graphical orbital/vector display.
Airspace / ADS-B	Nearby ADS-B aircraft reduced by the Gateway to a bounded target set with 3D perspective/vector display.

Area	What v1.0 provides
Task Manager	Three pages of truthful application/system diagnostics labeled LIVE/CALC/MAP/CONST/UNAVAILABLE.
Audio	UI sound cues, embedded startup march, and one background Music stream.
Operator tools	Global refresh, Local/Global scope, pause, Help, Diagnostics, Safe Idle, status/news-line advance.

3. Five-Minute Quick Start

Figure 2. Normal startup order for the accepted Windows/MAME workflow.

1. Start START_COMMAND_CENTER_GATEWAY.BAT in the integrated Gateway directory. Leave the Gateway console running.

2. Open the NABU Internet Adapter. Confirm its Local HomeBrew source points to the current client directory containing the accepted 000001.NABU, and confirm the Store path is D:\NABU Internet Adapter\Store.

3. For MAME, start the Adapter TCP program loader and then launch MAME 0.250 with the exact command in Chapter 25. For physical NABU, use the separately verified hardware deployment configuration.

4. Confirm the NABU loading screen, then the Command Center splash. Press any key to skip the splash if desired.

5. On the dashboard, confirm the footer/build identity before evaluating live data. For this manual the accepted identity is NCC-SPLASH-260826-LV2.

Physical NABU program loading — “PLEASE WAIT”.	Final v1.0 splash on physical NABU.

4. Dashboard Tour

The dashboard is a compact mission-console view. The Atomic Clock and global state live in the top header; the center grid contains five live-information modules plus NABU Task Manager. The bottom lines show status, controls, and BUILD_ID.

Figure 3. Annotated MAME dashboard development/acceptance capture. The screen layout is the owner-frozen presentation; the visible footer may predate the final LV2 build.

Marker	Region	Meaning
1	Header / Atomic Clock	Current date/time plus global operating state.
2	Global status	Music, sound, Local/Global, network and status indicators.
3	Earthquake	Recent seismic summary; opens Local/Global detail.
4	Space Weather	Kp / solar-wind status; opens trends.
5	Satellite / ISS	ISS state; opens orbital/vector view.
6	Airspace	Nearby aircraft summary; opens perspective target view.
7	Local Weather	Current conditions for the selected ZIP.

Marker	Region	Meaning
8	NABU Task Manager	Three pages of truthful diagnostics.
9	Footer / controls / BUILD_ID	Context keys, refresh/status line, and exact build identity.

	ATOMIC CLOCK PLACEMENT Atomic Clock is a persistent header/global service, not a sixth grid tile. The five live-data tiles plus NABU Task Manager occupy the dashboard grid.

PART II — Keyboard and Global Operating Model

5. Normal Operator Keys

Key	Context	Action
Arrow keys	Dashboard	Move the highlighted module selection.
ENTER	Dashboard	Open the selected module.
ESC or M	Most views	Return/back to the dashboard.
R	Dashboard or module	Queue an immediate manual global refresh.
Z	Dashboard only	Open five-digit ZIP / Location Profile entry.
L / G	Dashboard or module	Select Local or Global operating perspective where applicable.
P	Dashboard or module	Pause/resume scheduled display animation/update work; not a network-off switch.
X	Global	Toggle UI sound cues.
B	Global	Toggle background Music.
N	Dashboard only	Advance the bounded news/status line.
D	Global	Open Diagnostics.
H	Global	Open the compact single-page Help screen.
Q	Global	Enter Safe Idle view.
Left / Right	Task Manager	Change Task Manager page.
Left / Right	Target-aware detail	Select previous/next event, aircraft, or target where supported.
ENTER	Target-aware detail	Lock/unlock the current target where supported.

	KEYBOARD C The keyboard C key has no normal v1.0 operator action. Do not confuse it with AY sound channel C, which is used as Music voice 2.

	ENGINEERING KEYS J, U, and E are bounded qualification/service loops (100, 1000, and 5000 operations). They are deliberately excluded from the normal quick-reference card and appear only in the Advanced Service appendix.

6. Special Key Behavior

The NABU keyboard delivers special codes for the arrow keys. The production client maps Right F0, Left F1, Up F2, Down F3, ENTER 0D, ESC 1B, Backspace 08, and Delete 7F. You do not need to type those codes; they are recorded here so emulator/key-mapping problems can be distinguished from application problems.

Global keys are consumed before view-specific handling. This prevents a key intended for a global action from accidentally being interpreted as a module-specific target command.

7. Back, Selection, and Lock

Use arrows to move on the dashboard, ENTER to open a highlighted module, and ESC or M to return. In target-aware detail views, Left/Right cycles the bounded target list and ENTER toggles target lock. The footer on each screen is the authoritative immediate reminder.

	IF A KEY SEEMS IGNORED First verify focus is actually in the MAME window or that the physical NABU keyboard is responding. Then check the footer for the current screen. Do not assume a transport problem from an input symptom.

PART III — Startup, ZIP/Location, Refresh, and Status Vocabulary

8. Startup Splash

At program start, the Command Center presents its v1.0 identity before entering the dashboard. The final splash includes the product name, Version 1.0, the creator credit Derek Leger aka (Super_Derek), the 26 AUG 2026 date, and the native NABU presentation.

The splash plays a fixed embedded military/marching startup phrase. The tune is local to the NABU binary and does not depend on the Gateway or Store. The musical phrase is approximately eight seconds; the splash has a ten-second maximum. Pressing any key ends the splash immediately, silences the splash tune, consumes that keypress, and enters the dashboard without triggering the corresponding dashboard command.

Figure 4. Final NCC-SPLASH-260826-LV2 startup splash on physical NABU.

9. ZIP / Location Profile

Press Z from the dashboard to select the local operating location. The editor accepts exactly five decimal digits. Backspace or Delete removes one entered digit. ENTER submits a complete ZIP; ESC or M cancels and returns.

ZIP / Location Profile entry in MAME.	Another owner-captured ZIP-entry state showing the bounded five-digit editor.

1. From the dashboard, press Z.

2. Type exactly five digits.

3. Correct mistakes with Backspace or Delete.

4. Press ENTER to submit the Gateway request.

5. Wait for the location/weather response or press R once if you want an immediate global refresh.

6. Confirm the city/state label changes to the new accepted location before relying on Local-mode modules.

	CHANGING ZIP INVALIDATES THE PREVIOUS LOCATION PRESENTATION A new ZIP selection is not allowed to keep presenting the prior ZIP as if it were the newly accepted location. Until the new request is accepted, treat location-dependent state as pending/cached according to the screen status.

10. Automatic and Manual Refresh

Figure 5. Manual and automatic refresh are intentionally independent.

The v1.0 client schedules a global refresh approximately every 60 seconds. Pressing R queues an immediate manual global refresh but does not move or restart the automatic 60-second phase. Manual requests are prioritized when pending, and refresh work is serialized so overlapping foreground RetroNET operations do not run concurrently.

A successful refresh does not imply that every displayed number must change. Upstream data can legitimately be unchanged; in that case the same value can still be current.

11. Understanding Status Words

Status	Operator meaning	What to do
LIVE	A new/current valid state has been accepted.	Normal operation.
CACHED / CACHE	A previously accepted valid state is being retained.	Usually normal during unchanged data or temporary source trouble; check age/status.
STALE	Valid data is older than the module's freshness policy.	Press R once if needed; inspect Gateway/provider if it persists.
OFFLINE / OFF	No valid new result is currently available from that source/path.	Other modules may remain healthy; diagnose only the affected layer.
INVALID	A received record/state failed validation and was not accepted.	Preserve the screen/build evidence; check Gateway resource/log.
UNAVAILABLE	The program has no truthful supported value for that field.	Not a zero; the value is intentionally not claimed.
NOT CFG	The feature/source is not configured.	Configure the associated Gateway/provider rather than inventing a value.

PART IV — Atomic Clock and Local Weather

12. Atomic Clock

The Atomic Clock is always visible in the header. The Gateway maintains time from NIST; after a valid time record is accepted, the NABU advances the visible clock locally from its 60 Hz timebase instead of fetching the network continuously.

If the header says UNSYNC or obviously does not advance, press R once and watch the time/status. If Weather refreshes but the clock remains unsynchronized, capture the exact BUILD_ID and Gateway resource state rather than repeatedly pressing R.

Figure 6. Owner-captured Atomic Clock header detail from the development/acceptance corpus.

13. Local Weather

The Local Weather tile summarizes the selected ZIP. Open it with ENTER for current conditions and trends. The accepted detail view presents temperature, condition, wind, pressure, source/status information, and bounded history. Weather is tied to the currently accepted ZIP/location token.

Figure 7. Local Weather / Trends in MAME. This clean capture is used for text readability; its visible footer may predate final LV2.

13.1 Reading the Weather Detail

Area	Meaning
TEMP F	Current reduced Fahrenheit temperature.
WIND	Current reduced wind value/direction as provided by the Gateway presentation.
PRESS	Current reduced pressure.
STATUS	LIVE/CACHED/STALE/OFFLINE-style state for the accepted weather data.
SOURCE	Current provider label; production source is NWS.
History / trend graph	Bounded recent samples. It is a trend display, not a forecast model.
WX STAGE / age information	Compact freshness/history context used by the application.

13.2 Weather Alerts

The Weather module also receives a bounded NWS alert summary. v1.0 deliberately uses a stable cached alert presentation rather than the abandoned continuous scrolling crawl. The banner/summary updates when a new accepted alert record arrives or after refresh.

	WEATHER ALERT SCOPE Command Center is an informational display. For emergency decisions, use the issuing National Weather Service products and local official alerting channels.

13.3 Weather Trend Interpretation

Read the trend from older samples toward the newest sample according to the on-screen labels. A flat line can simply mean stable conditions. A missing/unchanged graph does not prove the Gateway is broken; check the status/source and whether a new accepted history record exists.

PART V — Earthquake and Space Weather

14. Earthquake Monitor

Earthquake uses USGS data reduced by the Gateway. The detail screen presents recent events in either Local or Global context, with magnitude, depth, age/time context, and a bounded region label. The native graphical terrain/vector view is an information visualization, not a geographic survey or emergency-command display.

Local Earthquake detail in MAME.	Global Earthquake detail in MAME.

Figure 8. Earthquake / Local Terrain running on the final accepted LV2 build on physical NABU.

14.1 Earthquake Controls

Key	Action
L	Select Local earthquake context.
G	Select Global earthquake context.
Left / Right	Move through retained events where the target list is available.
ENTER	Lock/unlock the selected event where supported.
R	Queue global manual refresh.
ESC or M	Return to dashboard.

15. Space Weather

Space Weather is reduced by the Gateway from NOAA Space Weather Prediction Center products. The accepted screen emphasizes a readable trend: Kp history, solar-wind history, a fixed Kp5 storm threshold, and a right telemetry rail with current severity/source state.

Clean MAME Space Weather / Trends capture.	Final accepted Space Weather view on physical NABU.

15.1 Reading the Trend

Display element	Meaning
KP 0-9 GREEN	Recent Kp history in the accepted readability presentation.
WIND KM/S YELLOW	Recent solar-wind speed history.
RED - KP5 STORM	Fixed Kp=5 reference threshold; it is a visual threshold line, not a separate live sensor.
STATUS / SEV	Compact current operating/severity state.
SOURCE SWPC	Production source identification.
OLDER → NEWEST	Direction of history chronology.

	SPACE-WEATHER INTERPRETATION This is a compact situational display. Use NOAA/SWPC's official products for operational warnings, forecasts, and technical interpretation.

PART VI — Satellite / ISS and Airspace / ADS-B

16. Satellite / ISS

The v1.0 Satellite module tracks the International Space Station. The production Gateway obtains ISS state from WhereTheISS and publishes a bounded record containing the values the NABU can use directly. The detail view preserves the owner-approved native orbital/vector visualization.

ISS orbital/vector view in MAME.	Final accepted ISS orbital/vector view on physical NABU.

16.1 What the ISS Screen Shows

- ISS identity and NORAD catalog identifier 25544.
- Current latitude and longitude from the accepted Gateway record.
- Altitude and velocity.
- Visibility/freshness information where present on the current screen.
- A graphical orbital/vector presentation derived from the accepted state.

16.2 Controls

Use Left/Right to move through target-aware state where the current screen exposes target selection, ENTER to lock/unlock where supported, R to refresh globally, and ESC or M to return.

	VISUALIZATION ACCURACY The orbital/vector display is a native situational visualization, not a precision orbital-navigation or prediction product.

17. Airspace / ADS-B

Airspace uses the Gateway to query ADSB.lol around the selected location, with a production radius of 100 nautical miles. The Gateway ranks/reduces the feed to a maximum of three aircraft for the NABU and prepares the bounded display coordinates. The native screen is a 3D perspective/vector visualization — intentionally not a conventional top-down radar scope.

Clean MAME Airspace perspective view.	Final accepted Airspace view on physical NABU.

17.1 Aircraft Information

- Callsign where the source provides one.
- ICAO aircraft identifier.
- Altitude.
- Speed.
- Heading.
- Relative/plot information prepared by the Gateway for the native display.

17.2 Selecting and Locking a Target

1. Open AIRSPACE from the dashboard.

2. Use Left/Right to move among the retained aircraft.

3. Press ENTER to lock the current target.

4. Read the callsign/ICAO and flight values in the telemetry rail.

5. Press ENTER again to unlock, or use Left/Right according to the footer to move to another target.

6. Press ESC or M to return.

	AIRSPACE LIMITATION NABU Command Center is not an aviation navigation, traffic-separation, collision-avoidance, ATC, or safety-of-life system. ADS-B data can be incomplete, delayed, filtered, or unavailable.

PART VII — Task Manager, Diagnostics, Help, and Safe Idle

18. NABU System / Task Manager

Task Manager is a truthful Command Center diagnostic instrument. It is not a modern operating-system task manager and does not invent CPU utilization, temperatures, voltages, or other unsupported hardware telemetry. Use Left/Right to move among its three pages and ESC or M to return.

Figure 9. Task Manager page 1/3 on the final accepted physical LV2 build.

Figure 10. Task Manager page 2/3 on the final accepted physical LV2 build.

Figure 11. Task Manager page 3/3 on the final accepted physical LV2 build.

18.1 Truth Labels

Label	Meaning	Examples
LIVE	Directly changing runtime/application state.	View, refresh state, counters, transport/application values.
CALC	Calculated from current program constants/state.	Network-buffer calculation, VRAM calculation.
MAP	Derived from linker/map evidence.	MAP D+B / 1104 B.
CONST	Fixed NABU/platform specification.	Z80A, ~3.58 MHz, 64K RAM, TMS9918A, 16K VRAM, AY-3-8910.
UNAVAILABLE	No truthful supported value is exposed.	USER DATA, LINK GAP.

18.2 Page 2 Resource Examples

Current v1.0 page 2 includes fixed or calculated values such as CPU Z80A, clock approximately 3.58 MHz, RAM 64K, VRAM 16K, TMS9918A, AY-3-8910, MAP D+B 1104 B, FONT C 315, VRAM CALC 13056, and explicit UNAVAILABLE fields where no valid runtime metric exists.

	646-BYTE MARGIN IS NOT FREE RAM Engineering documentation records only 646 bytes between the current BSS end and initial stack origin. That is an address-space/static-to-stack margin, not a user-facing 'free RAM' meter and not a Task Manager free-memory value.

19. Diagnostics

Press D to open the visible diagnostics screen. Use it to inspect current application/transport state, counters, view/state information, and the last visible diagnostic conditions that the production build exposes. Return with ESC or M.

The resource-constrained production build has the optional in-memory flight-recorder ring compiled out. The NABU client does not provide a persistent crash-log file. Gateway logging is separate on the modern host.

	WHEN REPORTING A PROBLEM Capture the Diagnostics screen if it is readable, the normal screen showing the failure, the visible BUILD_ID, whether you are in MAME or on physical NABU, and the matching Gateway/Adapter console state.

20. Help

Press H for the built-in Help screen. v1.0 already includes a compact single-page help view that lists core controls and the BUILD_ID. Return with ESC or M.

	WHY THE FULL MULTI-PAGE MANUAL IS NOT EMBEDDED A dense DOS-style multi-page About/Help system was evaluated after LV2 acceptance. The current program has only a 646-byte static-to-stack address-space margin, and verified Page Up/Page Down key codes were not part of the production input contract. The full manual therefore lives here rather than consuming the hardware-verified binary's remaining margin.

21. Safe Idle

Press Q to enter Safe Idle, a stable quiet/diagnostic view intended to stop normal dashboard/module animation and provide a controlled place to wait. It is not a power-off or operating-system exit function. Return according to the on-screen footer (normally M or ESC).

	DO NOT USE SAFE IDLE AS A NETWORK DISCONNECT Safe Idle is an application view, not a guarantee that every background service or the modern Gateway has stopped. If you need to stop network publication, stop the Gateway/Adapter at the host.

PART VIII — Sound, Music, and Startup Audio

22. UI Sound Cues

Press X to toggle Command Center UI sound cues. These short effects use the dedicated UI cue path. Sound cues are distinct from the Music toggle.

Control	Effect
X	UI sound cues ON/OFF.
B	Background Music ON/OFF.
C	No keyboard Music action in v1.0.

23. Background Music

Press B to toggle the v1.0 background Music stream. The Gateway acquires and converts a validated public-domain Mutopia MIDI source using the pinned Mido 1.3.3 host dependency. The NABU never parses raw MIDI; it receives a bounded AY-ready MU01 event segment and plays two tonal voices.

AY channel A is reserved for Command Center cues. AY channels B and C are available to the two Music voices. This hardware channel C has nothing to do with the unassigned keyboard C key.

Figure 12. MAME dashboard development capture showing MUSIC ON / CACHE status. Layout is representative; visible footer/build may predate final LV2.

23.1 Current v1.0 Music Limits

- Single curated stream; no genre-cycle control.

- Current production piece is the Mutopia public-domain Bach 'Aria Bist Du Bei Mir' source used by the Gateway conversion/cache path.
- Music playback is event-based AY synthesis, not streamed PCM/audio.
- The current compact arrangement has intentionally sparse passages; a few seconds of apparent silence can be musical content rather than a hang.
- R is a global data refresh request, not a 'restart song' button.

24. Splash March vs Background Music

The startup splash tune is embedded in the NABU binary and plays before the dashboard. It is separate from the Gateway-fed background Music service. Skipping the splash silences only the splash tune and transitions cleanly to normal application audio state.

PART IX — Everyday Operator Workflows

25. Morning Check — One Complete Pass

1. Boot Command Center and confirm the correct BUILD_ID.

2. Confirm the Atomic Clock advances.

3. Confirm the ZIP/city/state is the intended operating location; press Z if it is not.

4. Open Local Weather and review conditions/trends/alerts.

5. Open Earthquake and check Local; switch to Global with G if desired.

6. Open Space Weather and read Kp, solar-wind trend, and current severity/source.

7. Open Satellite / ISS and review current ISS state.

8. Open Airspace and inspect the retained nearby targets.

9. Open Task Manager if you want an application/system-health snapshot.

10. Toggle Music with B or UI sound cues with X according to preference.

26. Change Location and Recenter Local Services

1. Return to the dashboard.

2. Press Z.

3. Enter the new five-digit ZIP and press ENTER.

4. Wait for location acceptance. Confirm the new city/state label.

5. Press R once if you want to request an immediate global refresh.

6. Reopen Weather, Earthquake Local, and Airspace. These are the modules most directly affected by the selected local context.

27. Follow an Aircraft

1. Open AIRSPACE.

2. Read the aircraft count/range summary.

3. Use Left/Right to select a target.

4. Press ENTER to lock it.

5. Read callsign/ICAO, altitude, speed, and heading where available.

6. Press ENTER to unlock or choose another target.

7. Return with ESC or M.

28. Review a Seismic Event

1. Open EARTHQUAKE.

2. Press L for Local or G for Global.

3. Use Left/Right to select retained events where available.

4. Read magnitude, depth, age/time, and region.

5. Use the vector/terrain graphic as situational context only.

6. Return with ESC or M.

29. Check a Source That Looks Stuck

1. Read its status first: LIVE, CACHED, STALE, OFFLINE, or INVALID.

2. Press R once.

3. Wait for the foreground refresh to settle.

4. If the value is unchanged but status remains valid, remember that the provider may simply have no new value.

5. If only one module remains stale/offline, treat it as a provider/module issue first.

6. If all modules fail together, check the Gateway/Store layer.

PART X — MAME and Physical NABU Operation

30. Windows Paths Used by the Accepted Development Environment

Component	Verified project path
Client	<publication-repository>\client\phase3a_live_local_weather_v0_1\NABU_COMMAND_CENTER_LIVE_LOCAL_WEATHER_v0_1
Gateway	<publication-repository>\gateway\phase3a_live_local_weather_v0_1\NABU_COMMAND_CENTER_LOCAL_WEATHER_GATEWAY_v0_1
Store	D:\NABU Internet Adapter\Store
Internet Adapter	C:\Program Files (x86)\DJ Sures\NABU Internet Adapter\NABU-Internet-Adapter-84.exe
MAME 0.250	<MAME_ROOT> NABU 9938\mess.exe

31. Start the Integrated Gateway

1. Open the production Gateway directory.

2. Run START_COMMAND_CENTER_GATEWAY.BAT.

3. The launcher prefers py -3 and falls back to python. It verifies the installed mido distribution is exactly 1.3.3 before importing the Music gateway path.

4. Leave the console running while using live Command Center modules.

5. Normal publication messages show the modern side is continuing to maintain Store resources.

Figure 13. Owner-captured integrated Gateway console during development/acceptance. Use the current launcher, not this screenshot's transient values, as authority.

32. Start the NABU Internet Adapter

1. Open NABU-Internet-Adapter-84.exe.

2. Under Settings → Source, enable the verified Local HomeBrew source and point it to the current client folder containing 000001.NABU.

3. Confirm the Store root is the intended D:\NABU Internet Adapter\Store.

4. For MAME program loading, start the TCP listener used by the loader path on port 5816.

5. If 000001.NABU has been replaced, stop/start the program loader so the Adapter listener/cache cannot serve a stale program.

6. Do not restart the Adapter just because a Store telemetry record changed; program-loader caching and runtime Store freshness are different mechanisms.

Figure 14. NABU Internet Adapter interface captured during project verification. This development capture also contains an 'end of stream' diagnostic later discussed in Troubleshooting; use the current settings/procedure, not the transient log lines, as authority.

33. Run Command Center in MAME 0.250

From the verified MAME directory, the current normal command is:

mess nabupc -window -kbd nabu_hle -hcca null_modem -bitb socket.127.0.0.1:5816

After MAME opens, pass any normal MAME information screen, give the emulator window keyboard focus, watch the NABU load/splash sequence, then verify the visible BUILD_ID before acceptance testing.

	PORT ROLES TCP 5816 is the MAME program-loader path. Runtime Command Center Store traffic uses the Internet Adapter's RetroNET Store service (historically port 5815 in the verified environment). Do not treat 5816 as an application data socket.

34. Running on a Physical NABU

The exact NCC-SPLASH-260826-LV2 binary was owner-verified running on physical NABU hardware. Physical operation therefore has direct runtime evidence. However, this manual deliberately does not invent the exact USB/RS-422 adapter model, COM port, cable/pinout, baud/serial fields, or electrical details needed for a third party to recreate the connection.

1. Use the separately owner-verified physical deployment configuration and cable/interface record.

2. Point the Internet Adapter at the exact accepted 000001.NABU.

3. Start the documented physical listener/mode rather than the MAME TCP loader.

4. Power/reset the NABU according to the verified hardware procedure.

5. Confirm the loading screen, splash, and exact BUILD_ID.

6. Exercise the normal modules and compare behavior with the MAME-accepted baseline.

Physical NABU loading the accepted application.	Final LV2 live application running on physical NABU.

PART XI — Troubleshooting and Recovery

Figure 15. Start by locating the failed layer; do not rewrite or reconfigure unrelated layers.

35. Program Does Not Load

If MAME/NABU never reaches the Command Center splash or shows an Adapter/program-load failure, the first concern is the program-loading path, not Weather, ADS-B, or other modules.

1. Confirm the Internet Adapter is running.

2. Confirm the Local HomeBrew source points to the intended client directory.

3. Confirm the intended 000001.NABU is present.

4. Restart/invalidate the Adapter program-loader listener/cache if the binary was replaced.

5. Confirm MAME is using the exact 5816 program-load command.

6. Capture the Adapter log and any NABU error screen before changing source.

Figure 16. Historical development example of an ADAPTOR FAILURE screen. This is a troubleshooting illustration, not a final LV2 normal screen.

36. Dashboard Loads but Live Data Fails

If the application and clock UI are alive but several live modules remain OFFLINE/STALE together, check the integrated Gateway and Store publication first.

1. Confirm START_COMMAND_CENTER_GATEWAY.BAT is still running.

2. Check that Store resources are being published under D:\NABU Internet Adapter\Store.

3. Press R once and allow the refresh to finish.

4. If every module still fails, capture Gateway console output and Adapter state.

5. Do not assume the Golden Transport has regressed without evidence from the bounded read/validation layer.

37. Only One Module Is Offline or Stale

The providers are isolated. A failing USGS, SWPC, ISS, ADS-B, NWS, or Music source should not be diagnosed as a whole-system network failure solely because that module is unavailable. Preserve last-valid state and diagnose the affected provider/resource.

Symptom	First operator interpretation
Weather stale; other modules live	Weather/NWS/location resource path first.
Earthquake stale; others live	USGS/source-specific path first.
Space Weather stale; others live	NOAA/SWPC path first.
ISS stale; others live	WhereTheISS/source-specific path first.
Airspace stale; others live	ADSB.lol/current ZIP/radius query path first.
Music silent; dashboard live	Check B state, Music status/cache, Gateway dependency/source; sparse musical passages are possible.

38. ZIP Does Not Resolve

1. Confirm the editor contains exactly five digits.

2. Submit with ENTER.

3. Read the location/profile status.

4. Press R once if you want immediate refresh.

5. If the old city/state persists with an error/pending state, do not assume it represents the newly entered ZIP.

6. Check Gateway location/weather messages.

39. Screen Corruption

The final Weather detail renderer includes the correction for a real stack/buffer corruption defect found during LV1 testing. If any comparable corruption appears in LV2 or a later build, treat the screen as evidence rather than continuing to navigate aggressively.

1. Photograph or capture the corrupted screen.

2. Record BUILD_ID and the module/action that preceded it.

3. Capture Gateway/Adapter state if networking was active.

4. Reset/reload the known accepted binary.

5. Report whether the corruption is repeatable from the same action.

Figure 17. Historical pre-LV2 Weather corruption capture retained as defect evidence. It is not expected final behavior.

40. Adapter Reports 'end of stream'

Treat 'end of stream' as a connection/session symptom, not a root-cause diagnosis. During development it appeared alongside a Weather renderer corruption event, but the renderer stack defect and the Adapter message were not proven to have a direct causal relationship. Capture both sides and change one principal variable at a time.

41. Music Sounds Sparse or Stops Briefly

The current Music arrangement is intentionally compact. The accepted track contains sparse monophonic tail passages and a short final silence. If the application remains responsive and Music state continues normally, a quiet passage is not evidence of a hang. If Music never resumes, toggle B once, inspect Music status/cache, and then check the Gateway.

42. Wrong BUILD_ID

A wrong BUILD_ID is an identity/cache problem until proven otherwise. Stop evaluating module behavior against the wrong binary. Reconfirm the intended 000001.NABU, invalidate/restart the Adapter program loader, reload, and verify the footer again.

PART XII — Data Sources, Limits, Maintenance, and Service Controls

43. Production Information Sources

Feature	Production source	What the NABU receives
Atomic Clock	NIST Internet Time Service	Compact time state; NABU advances locally.
Weather / location	NWS API + U.S. Census ZCTA coordinates	Bounded current conditions, location, and history records.

Feature	Production source	What the NABU receives
Weather alerts	NWS Alerts API	Bounded count/severity/text summary.
Earthquake	USGS real-time GeoJSON summary feed	Reduced Local/Global event set.
Space Weather	NOAA / SWPC	Reduced current metrics and bounded history.
Satellite / ISS	WhereTheISS, satellite 25544	ISS position/velocity state.
Airspace / ADS-B	ADSB.lol point API, 100 NM radius	Up to three reduced aircraft targets.
Music	Mutopia piece 257 + Mido 1.3.3 conversion	Validated bounded AY-ready event data.

44. What Command Center Does Not Claim

Display	Important limit
Weather / alerts	Informational. Use official weather/emergency channels for protective decisions.
Earthquake	Informational summary, not an emergency-command or engineering hazard system.
Space Weather	Compact situational summary; use NOAA/SWPC for operational interpretation.
Satellite / ISS	Visualization, not precision orbital navigation or pass-prediction authority.
Airspace	Not ATC, navigation, collision avoidance, or a safety-of-life system.
Task Manager	Not CPU utilization, thermal, voltage, or unrestricted free-memory telemetry.
Diagnostics	No persistent NABU crash-log file in the current resource-constrained build.

45. Maintaining the NABU Program Artifact

1. Preserve the accepted 000001.NABU and its SHA-256 before replacing it.

2. Install a newer version only from a known release package.

3. Restart/invalidate the Adapter program-loader listener/cache after replacing the program file.

4. Reload the NABU/MAME program.

5. Confirm the new visible BUILD_ID before accepting the update.

6. Keep the older accepted binary available for rollback.

46. Maintaining the Gateway

1. Preserve command_center.ini and any release-specific configuration.

2. Use START_COMMAND_CENTER_GATEWAY.BAT as the integrated entry point.

3. Keep the pinned requirements-music.txt dependency available; v1.0 requires mido==1.3.3 for Music conversion/import.

4. Do not place API keys or secrets inside the NABU binary.

5. After a Gateway update, verify normal Store publication before blaming the NABU client for missing data.

47. Advanced / Service Qualification Keys

	NOT REQUIRED FOR NORMAL OPERATION The following keys exist for bounded engineering qualification. Use them only when following a service/acceptance procedure and record the result.

Key	Bounded service action
J	Run the 100-operation qualification loop.
U	Run the 1000-operation qualification loop.
E	Run the 5000-operation qualification loop.

APPENDIX A — One-Page Keyboard Reference

Key	Normal v1.0 action
↑↓←→	Move dashboard selection; Left/Right also select pages/targets where the footer says so.
ENTER	Open selection; lock/unlock supported detail targets.
ESC / M	Back / return to dashboard.
R	Immediate manual global refresh; automatic 60-second schedule remains independent.
Z	Dashboard-only five-digit ZIP editor.
Backspace / Delete	Delete one ZIP digit while editing.
L / G	Local / Global perspective.
P	Pause/resume scheduled display animation/update work.
X	UI sound cues ON/OFF.
B	Background Music ON/OFF.
N	Dashboard-only status/news-line advance.
D	Diagnostics.
H	Compact Help.
Q	Safe Idle.
C	No normal v1.0 assignment.

APPENDIX B — Status Quick Reference

State	Meaning
LIVE	New/current valid state accepted.
CACHED / CACHE	Previously accepted valid state retained.
STALE	Valid but older than freshness policy.
OFFLINE / OFF	No valid new source result currently available.
INVALID	Received state failed validation.

State	Meaning
UNAVAILABLE	No truthful supported value exists.
NOT CFG	Feature/source not configured.
SEV NOM	Compact nominal severity indication.
SEV ADV	Compact advisory-level severity indication where the module uses it.

APPENDIX C — Module Quick Reference

Module / service	Open / control	Primary source	Key user content
Atomic Clock	Always visible; R refresh	NIST	Time/date, sync/cache state.
Local Weather	Select tile → ENTER	NWS	Temp, condition, wind, pressure, trends, alert summary.
Earthquake	Select → ENTER; L/G; Left/Right; ENTER lock	USGS	Magnitude, depth, age, region, Local/Global vector view.
Space Weather	Select → ENTER	NOAA/SWPC	Kp, solar wind, trend, threshold, severity/source.
Satellite / ISS	Select → ENTER; target controls where shown	WhereTheISS	ISS 25544, lat/lon, altitude, velocity, orbital vector.
Airspace / ADS-B	Select → ENTER; Left/Right; ENTER lock	ADSB.lol	Up to 3 aircraft, callsign/ICAO, altitude/speed/heading, perspective plot.
NABU Task Manager	Select → ENTER; Left/Right pages	Native client	LIVE/CALC/MAP/CONST/UNAVAILABLE diagnostics.

APPENDIX D — Startup Checklist

- Gateway: START_COMMAND_CENTER_GATEWAY.BAT running without fatal dependency error.
- Store: D:\NABU Internet Adapter\Store is the intended publication root.
- Adapter: intended Local HomeBrew source selected.
- Adapter: program loader restarted after any 000001.NABU replacement.
- MAME: exact normal command uses nabupc, nabu_hle, null_modem, and socket.127.0.0.1:5816.
- NABU: PLEASE WAIT → splash → dashboard.
- Identity: NCC-SPLASH-260826-LV2 for this accepted manual baseline.
- Location: correct ZIP/city/state.
- Clock: advancing.
- Modules: status labels make sense; do not require every value to change on every refresh.

APPENDIX E — Troubleshooting Evidence Packet

When a repeatable problem needs engineering review, collect the smallest complete evidence packet:

- Exact BUILD_ID visible on the screen.
- MAME or physical NABU.

- Photograph/screenshot of the failed screen.
- Last key/action before the symptom.
- Whether the symptom repeats from a clean reload.
- Gateway console excerpt at the same time.
- Internet Adapter log/state when program loading or transport is implicated.
- Which other modules remained healthy.
- For a stop/hang under debugger: PC, SP, surrounding disassembly, registers/flags, symbol/map location, interrupt state, and whether MAME is paused.

APPENDIX F — Build Identity and Acceptance Record

Item	Accepted LV2 value/status
BUILD_ID	NCC-SPLASH-260826-LV2
000001.bin	59,501 bytes
000001.NABU	59,501 bytes
Binary identity	Byte-identical
SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
Compilation / link	VERIFIED
MAME execution	OWNER VERIFIED
Weather maximized-view correction	VERIFIED
Splash	VERIFIED
Task Manager	VERIFIED
Integrated modules	VERIFIED
Physical transfer/execution	OWNER VERIFIED
Physical GUI/runtime operation	OWNER VERIFIED
Final LV2 Git commit/tag	PENDING / NONE at this manual baseline

APPENDIX G — Screenshot / Photograph Provenance

The manual intentionally uses real owner-supplied development and acceptance captures. Final physical photographs are preferred where they prove the accepted LV2 build; clean MAME captures are used where they make small text easier to read. Older MAME captures are labeled as development/acceptance illustrations rather than presented as final binary identity.

Original file	Environment	Identity	Use in manual
1000074246.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Airspace / ADS-B final accepted view
1000074247.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Task Manager page 1/3
1000074248.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Task Manager page 2/3
1000074250.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Task Manager page 3/3

Original file	Environment	Identity	Use in manual
1000074251.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Space Weather / Trends
1000074252.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Earthquake / Local Terrain
1000074253.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Satellite / ISS Orbital Vector
1000074254.jpg	Physical NABU	accepted LV2 load sequence	PLEASE WAIT
1000074255.jpg	Physical NABU	NCC-SPLASH-260826-LV2	Version 1.0 splash
0a77a710-4e22-4361-b455-041ddb1e225e.png	MAME	development/acceptance	Dashboard
43f8f143-5aaa-4d7a-85b3-adbef2fb67bd.png	MAME	development/acceptance	ZIP / Location Profile
76387ad0-ca93-4759-b6f1-e7d0ba4d3470.png	MAME	development/acceptance	Local Weather / Trends
602c7269-4923-4b8d-ab92-b5a61460014f.png	MAME	development/acceptance	Earthquake Local
7312aa24-e21e-41e8-9d1d-a937e7a928a3.png	MAME	development/acceptance	Earthquake Global
3dc59e0c-45a9-48c8-b5cc-1fb2c38f18fb.png	MAME	development/acceptance	Space Weather
5c1c7d24-95ec-4228-be9e-e1aecbffe7d6.png	MAME	development/acceptance	Satellite / ISS
0b45d4ed-0816-4627-9e90-6248ed99bbf6.png	MAME	development/acceptance	Airspace / ADS-B
0d79a73c-a302-4ed4-9034-bde74ff1aab7.png	MAME	Music development/acceptance	Dashboard with Music ON / CACHE
79045ddf-4934-44f0-9ba8-9346d80f02b6.png	Windows host	project verification	NABU Internet Adapter
7fe2031f-e968-433c-8cfd-884344c50326.png	Windows host	project verification	Integrated Gateway console
9a707a49-694d-4a08-83cf-f4f1741a8070.png	MAME	historical defect	ADAPTOR FAILURE example
22ee1d85-3f4d-4b8a-a891-d22d272fb0e4.png	MAME	historical defect	Pre-LV2 Weather corruption evidence

APPENDIX H — Documentation and Source Basis

Reference	Role
NABU Command Center Master Production Engineering White Paper v1.1	Current consolidated production architecture, controls, resource contracts, Gateway/provider map, MAME/physical status, and release boundary.
NABU Command Center Codex Technical Review and Update Recommendations, 26 Aug 2026	Independent read-only source/Gateway/map/Git audit used to correct the master paper and preserve source-level truth.
Current NCC-SPLASH-260826-LV2 main.c / BUILD.BAT / map/symbol/binaries	Implementation authority for controls, views, scheduler, splash, Help, memory/resource state, and artifact identity.
Agent_Instructions_SuperPrompt.txt	Evidence doctrine, verified NABU workflow, and physical/network non-invention rules.

Reference	Role
Original NABU Technical Specifications / Technical Manual	Platform and keyboard reference.
Owner MAME and physical-NABU acceptance photographs, 26 Aug 2026	Direct runtime evidence for the integrated accepted LV2 build.

Provider Credits / Official Source Families

NIST Internet Time Service; National Weather Service api.weather.gov and Alerts API; U.S. Census ZCTA coordinates used by the Gateway for ZIP-centered location work; U.S. Geological Survey real-time earthquake feeds; NOAA Space Weather Prediction Center; WhereTheISS; ADSB.lol; and the Mutopia Project. Provider availability, policies, schemas, and endpoints can change independently of the NABU client.

APPENDIX I — Known v1.0 Limitations

- Final LV2 Git commit/tag freeze is pending; the exact runtime binary is owner accepted but repository freeze is a separate gate.
- The full multi-page in-program About/Help manual is not embedded; H opens a compact single-page Help view.
- The keyboard C key does not cycle Music genres; v1.0 has one curated Music stream.
- The production diagnostic flight-recorder ring is compiled out; no persistent NABU crash log is exposed.
- Physical NABU execution is verified, but the complete third-party hardware interconnect manifest remains to be finalized; do not guess wiring/COM/serial details.
- Live provider availability can change. Last-valid/cache behavior is designed to keep unrelated modules usable.
- All public-data visualizations are informational, not safety-of-life or navigation products.

END OF USER MANUAL

NABU COMMAND CENTER v1.0 • NCC-SPLASH-260826-LV2